

Introduction to Statistics

Data, Estimation, and Inference

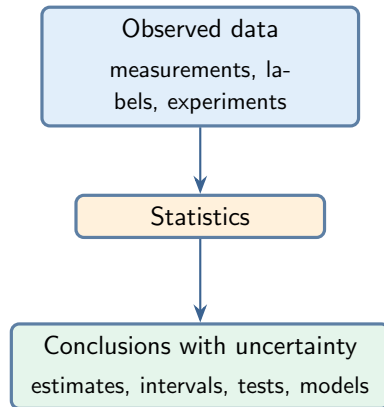
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AI and AIS Summer School July/Aug 2026

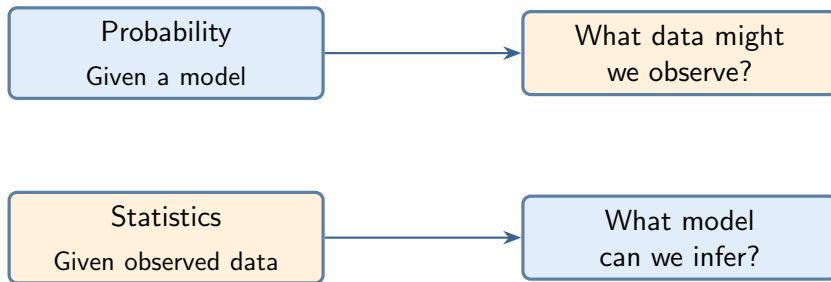
Why Statistics Matters in AI and Machine Learning

- Machine learning starts from observed data.
- Data are finite, noisy, and often biased.
- We estimate unknown quantities from samples.
- We need to know whether apparent patterns are reliable.

Statistics gives us tools for learning from data under uncertainty.



Probability and Statistics: The Direction Matters



In this part, we focus on **statistics**: how to infer structure from data.

An Example: Measurements from a Process

Suppose we repeatedly read a temperature sensor.

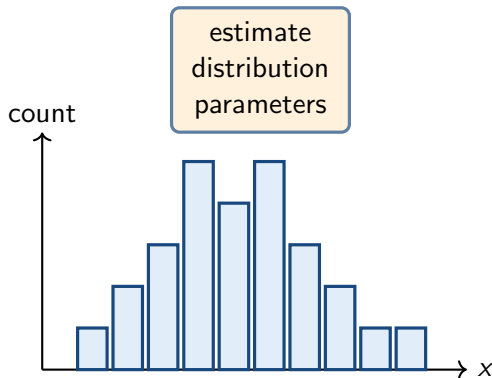
Observed sample

$$x_1, x_2, \dots, x_n$$

- Each value is one measurement.
- The underlying distribution is unknown.
- We want to estimate its parameters.

Example questions

What is the mean temperature? How variable is it? Does a new batch look compatible with this distribution?



Population and Sample

Population

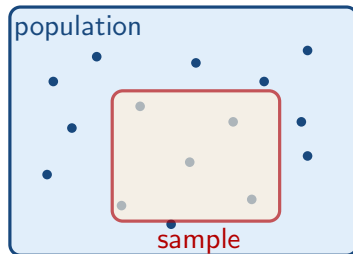
The full collection we would like to understand.

- all future operating points
- all possible products possible users of a product

Sample

The data we actually observe.

$$\{x_1, x_2, \dots, x_n\}$$



Statistics uses the sample to say something about the population.

Data, Parameters, and Statistics

Parameter

A fixed but usually unknown property of the population.

$$\mu, \sigma^2, \rho, \theta$$

Examples: true mean, true variance, true model coefficient.

Statistic

A quantity computed from observed data.

$$\bar{x}, s^2, r, \hat{\theta}$$

Examples: sample mean, sample variance, sample correlation, fitted coefficient.

Key idea

A statistic is observable. A parameter is what we want to learn about.

Descriptive Statistics: First Summary of Data

Descriptive statistics summarise a dataset before building a model.

Location

$$\text{mean: } \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

median: middle value after sorting

Shape

- symmetric or skewed;
- concentrated or spread out;
- one cluster or several clusters;
- outliers or unusual points.

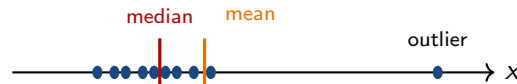
Spread

range, interquartile range, s

Mean, Median, Mode, and Outliers

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- The **mean** uses all values.
- The **median** is the middle value after sorting.
- The **mode** is the most frequently occurring value.
- Outliers can move the mean strongly.



No mode: each
observed value
occurs only once

A mode may not be unique

A data set may have one mode, several modes, or no mode.

Here, every observation is different, so there is no mode.

The True Distribution and the Data We Observe

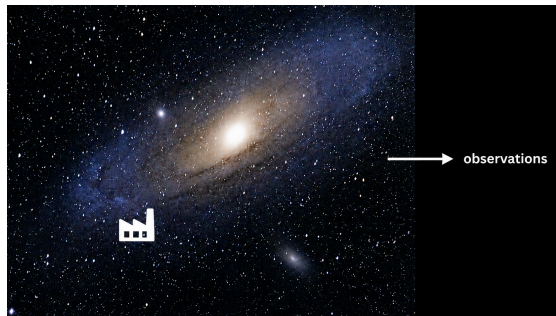
If we have the means—sensors, detectors, eyes, ears, etc.—we can observe random processes in the universe, whether natural or man-made:

$$X_1, X_2, \dots, X_n.$$

These observations come from some underlying probability distribution of the data-generating process, which we denote by

$$P_X \quad \text{or} \quad F.$$

This is the **true distribution**.

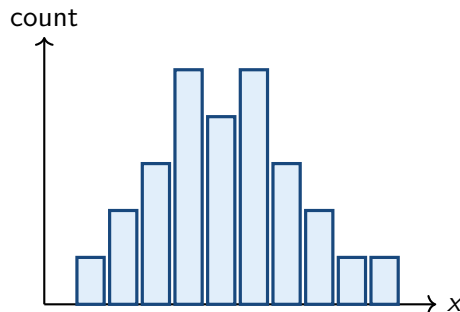


The true distribution exists, but it is **unknown** to us. Sometimes we may be able to model it fairly accurately from first principles, but it is not something that can be observed directly.

Histograms: Seeing the Empirical Distribution

A histogram shows how often values fall into intervals.

- It is a data-based picture of a distribution.
- It helps reveal shape, spread, and outliers.
- The choice of bin width affects the appearance.



Sample Variance and Standard Deviation

The sample variance measures spread around the sample mean:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

The sample standard deviation is

$$s = \sqrt{s^2}.$$

Why divide by $n - 1$?

The sample mean $\bar{x}$ is estimated from the same data. Dividing by $n - 1$ corrects the typical underestimation of population variance.

Standard deviation is easier to interpret because it has the same units as the data.

Standardisation: Putting Variables on a Common Scale

Features can have very different units and magnitudes.

$$z_i = \frac{x_i - \bar{x}}{s}.$$

Before standardisation

- temperature in degrees C;
- pressure in Pa;
- flow rate in kg/s.

After standardisation

- mean approximately 0;
- standard deviation approximately 1;
- variables are more comparable.

ML connection

Standardisation often improves optimisation and prevents one large-scale feature from dominating the model.

Data Matrix: Observations and Features

For machine learning, data are often organised as a matrix:

$$\mathbf{X} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}.$$

Rows

Observations or samples:

$$i = 1, \dots, n.$$

Columns

Variables or features:

$$j = 1, \dots, p.$$

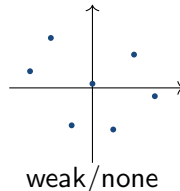
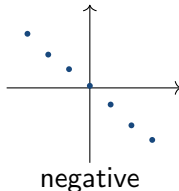
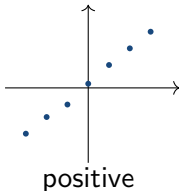
Covariance and Correlation in Data

Sample covariance measures whether two variables move together:

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}).$$

Sample correlation normalises covariance:

$$r_{xy} = \frac{s_{xy}}{s_x s_y}.$$

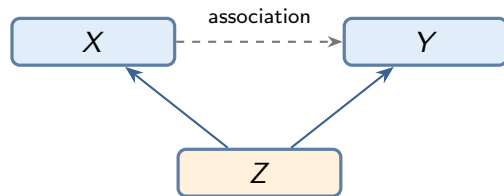


Correlation Is Not Causation!

A correlation can be useful, but it does not prove that one variable causes the other.

Possible explanations

- X affects Y .
- Y affects X .
- A third variable affects both.
- The pattern is accidental in a small sample.



Controlled experiments and causal reasoning are needed for causal claims.

Estimation: Learning Unknown Quantities

Statistics often asks:

What is the unknown parameter θ ?

We estimate it from data:

$$\hat{\theta} = \text{function of } x_1, x_2, \dots, x_n.$$

Example

$$\theta = \mu, \quad \hat{\theta} = \bar{x}.$$

The sample mean estimates the population mean.

Distribution example

$$\theta = (\mu, \sigma^2), \quad \hat{\theta} = (\bar{x}, s^2).$$

We estimate the centre and spread of a distribution from data.

Sampling Variability

If we collect a different sample, the statistic will usually change.

sample 1

$\bar{x}_1$

sample 2

$\bar{x}_2$

sample 3

$\bar{x}_3$

sample 4

$\bar{x}_4$

Key idea

A statistic is random before the data are collected. Its distribution across repeated samples is called a **sampling distribution**.

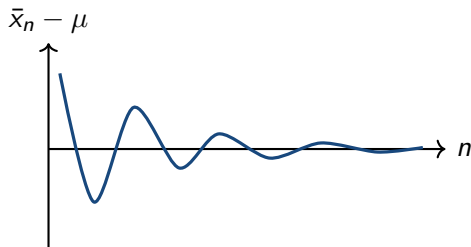
This is the basis for standard errors, confidence intervals, and hypothesis tests.

The Law of Large Numbers

As the sample size grows, the sample mean tends to get closer to the population mean.

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \quad \implies \quad \bar{X}_n \approx \mu \quad \text{for large } n.$$

- Small samples can be misleading.
- Larger samples usually give more stable estimates.
- More data helps, but only if the data are relevant and representative.



The Central Limit Theorem

For many situations, the sample mean is approximately normally distributed when n is large.

$$\bar{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right).$$

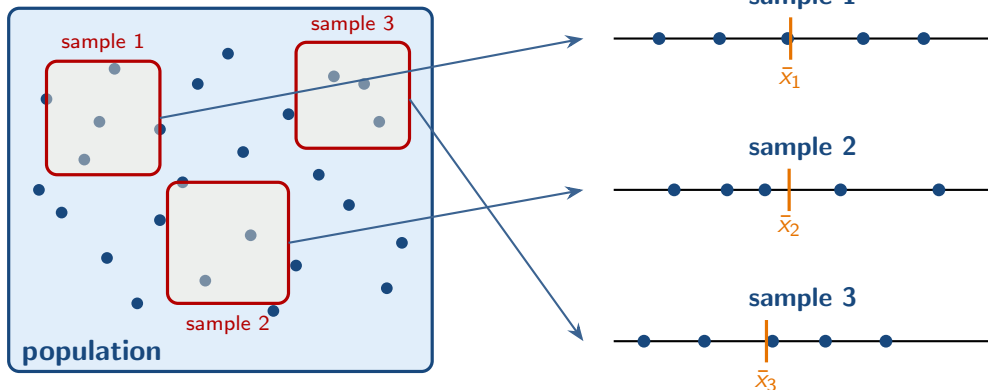
Equivalently,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx \mathcal{N}(0, 1).$$

Why this is important

Even if the raw data are not Gaussian, averages often behave approximately like Gaussian random variables.

Central Limit Theorem: Many Samples



The sample means $\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots$ behave approximately as draws from a Gaussian distribution. This has important uses e.g. when working with batches of data in machine learning applications.

Standard Error: Uncertainty of an Estimate

The standard deviation of an estimator is called its **standard error**.

For the sample mean,

$$\text{STDE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Since the population standard deviation σ is usually unknown, we estimate it using the sample standard deviation s :

$$\widehat{\text{STDE}}(\bar{X}) = \frac{s}{\sqrt{n}}.$$

Interpretation

The standard deviation describes variation among the observations.

The standard error describes how much the estimated mean would vary across repeated samples.

Confidence Intervals: Estimate Plus Uncertainty

A confidence interval gives a range of plausible values for an unknown parameter.

estimate $\pm$ margin of error.

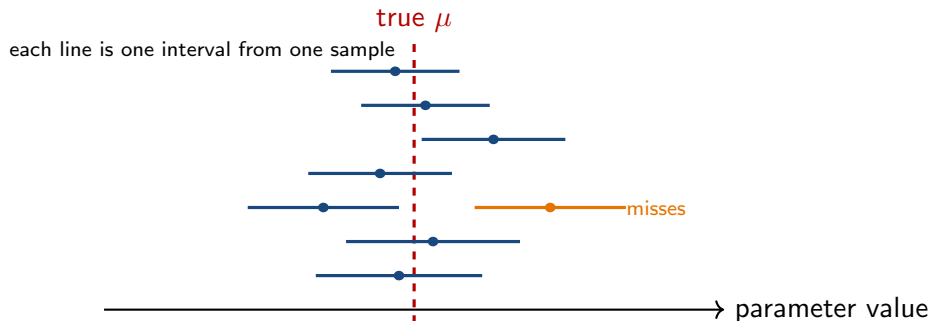
For a large-sample interval for a mean:

$$\bar{x} \pm 1.96 \frac{s}{\sqrt{n}}.$$

Interpretation

A 95% confidence interval comes from a procedure that captures the true parameter in about 95% of repeated samples.

Confidence Intervals: Picture



A confidence interval is random because it depends on the sample.

Why Descriptive Statistics Are Not Enough

Suppose we observe data

$$x_1, x_2, \dots, x_n.$$

Descriptive statistics can tell us:

$$\bar{x} = 12.4, \quad s = 2.1.$$

But they do not directly answer questions such as:

$$P(X > 15) = ?$$

$$P(10 \leq X \leq 13) = ?$$

From Descriptive to Inferential Statistics

Descriptive statistics

Summarise the data we have observed:

$\bar{x}$, s , median, histogram.

Inferential statistics

Use observed data to reason about the process that generated the data:

$$x_1, \dots, x_n \longrightarrow \hat{\theta} \longrightarrow P(X_{\text{new}} \in A \mid \hat{\theta}).$$

Observed data $\longrightarrow$ model of uncertainty $\longrightarrow$ predictions about future observations.

A Distribution as a Model

A PMF or PDF can be used as a model for how observations are generated.

Discrete data

$$P(X = x) = p(x)$$

Example:

X = number of defects.

Continuous data

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Example:

X = measurement error.

Once we have a distributional model, we can compute probabilities of new observations.

Model Type and Parameters Control the Shape of the Model

The parameters determine the behaviour of the distribution.

- For a Gaussian distribution, μ controls location and σ controls spread.
- For an exponential distribution, λ controls the rate of decay.
- For a Poisson distribution, λ controls the expected count.

The model form also matters:

Gaussian shape $\neq$ Exponential shape $\neq$ Poisson shape.

Important message

Inference is not only about estimating numbers.
It is also about choosing a reasonable model form.

Examples of Parametric Distribution Models

Gaussian

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right)$$

$$x \in \mathbb{R}$$

$$\theta = \begin{bmatrix} \mu \\ \sigma \end{bmatrix}$$

- μ : location
- $\sigma > 0$: spread

Exponential

$$f(x; \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$\theta = \lambda, \quad \lambda > 0$$

$$\mathbb{E}[X] = \frac{1}{\lambda}$$

- λ : decay rate
- larger λ gives shorter waiting times

Likelihood: How Well Does a Model Explain the Data?

Suppose the statistical model is a family of probability distributions $\{P_\theta : \theta \in \Theta\}$, where each value of θ defines a different distribution.

After observing data $x_1, \dots, x_n$, the likelihood treats the observed data as fixed and varies θ :

$$L(\theta) = \prod_{i=1}^n p_\theta(x_i),$$

where p_θ denotes the model PMF or PDF. It is often easier to work with the log-likelihood:

$$\ell(\theta) = \log L(\theta) = \sum_{i=1}^n \log p_\theta(x_i).$$

- The logarithm converts products into sums, which are usually easier to differentiate and optimise.
- Since the logarithm is strictly increasing, maximising $\ell(\theta)$ gives the same estimate as maximising $L(\theta)$.

Maximum Likelihood Estimation

Maximum likelihood chooses the parameter value that makes the observed data most probable.

$$\hat{\theta}_{\text{MLE}} = \arg \max_{\theta} L(\theta) = \arg \max_{\theta} \ell(\theta).$$

Example: biased coin

If we observe k heads in n tosses, then

$$\hat{p}_{\text{MLE}} = \frac{k}{n}.$$

Interpretation

MLE says: choose the probability of heads equal to the observed fraction of heads.

Exercise: Estimating an Exponential Distribution

Suppose X is the waiting time until an event occurs. We observe the following waiting times:

0.6, 1.1, 1.4, 1.8, 2.0, 2.4, 2.7, 4.0.

Modelling assumption

Assume that the waiting times are independent observations from an exponential distribution:

$$X \sim \text{Exp}(\lambda), \quad f(x; \lambda) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

Questions

- 1 Estimate the unknown rate parameter λ .
- 2 Use the fitted distribution to estimate

$$P(1 \leq X \leq 3).$$

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- 2 Estimate

$$P(1 \leq X \leq 3)$$

using the fitted model.

Step 1: Construct the Likelihood

For independent observations $x_1, \dots, x_n$,

$$\begin{aligned} L(\lambda) &= \prod_{i=1}^n f(x_i; \lambda) \\ &= \prod_{i=1}^n \lambda e^{-\lambda x_i} \\ &= \lambda^n \exp \left(-\lambda \sum_{i=1}^n x_i \right). \end{aligned}$$

It is easier to maximise the log-likelihood:

$$\begin{aligned} \ell(\lambda) &= \log L(\lambda) \\ &= n \log \lambda - \lambda \sum_{i=1}^n x_i. \end{aligned}$$

Why use the log-likelihood?

Step 2: Estimate the Rate Parameter

Differentiate the log-likelihood and set the derivative equal to zero:

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0$$
$$\implies \hat{\lambda} = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}}.$$

For the observed data,

$$\sum_{i=1}^8 x_i = 0.6 + 1.1 + 1.4 + 1.8 + 2.0 + 2.4 + 2.7 + 4.0$$
$$= 16, \quad \bar{x} = \frac{16}{8} = 2.$$

Therefore,

$$\hat{\lambda} = \frac{1}{\bar{x}} = \frac{1}{2} = 0.5$$

Step 3: The Fitted Distribution

Substituting $\hat{\lambda} = 0.5$, the fitted model is

$$X \sim \text{Exp}(0.5).$$

Fitted PDF

$$\hat{f}(x) = 0.5e^{-0.5x}, \quad x \geq 0.$$

For an exponential distribution,

$$\mathbb{E}[X] = \frac{1}{\lambda}.$$

The expected waiting time under the fitted model is therefore

$$\mathbb{E}_{\hat{\lambda}}[X] = \frac{1}{\hat{\lambda}} = \frac{1}{0.5} = 2.$$

Fitted CDF

$$\hat{F}(x) = P(X \leq x) = 1 - e^{-0.5x}.$$

For the exponential distribution PDF is,

$$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

Therefore, its CDF is

$$\begin{aligned} F(x) &= \int_0^x \lambda e^{-\lambda u} du \\ &= 1 - e^{-\lambda x}. \end{aligned}$$

Step 4: Probability of a Value in a Range

Using the fitted CDF,

$$P(1 \leq X \leq 3) = \hat{F}(3) - \hat{F}(1).$$

Since $\hat{F}(x) = 1 - e^{-0.5x}$,

$$\begin{aligned} P(1 \leq X \leq 3) &= \left(1 - e^{-0.5(3)}\right) - \left(1 - e^{-0.5(1)}\right) \\ &= e^{-0.5} - e^{-1.5} \\ &\approx 0.6065 - 0.2231 \\ &= \boxed{0.3834}. \end{aligned}$$

Interpretation

According to the fitted exponential model, a new waiting time has approximately a 38.3% probability of falling between 1 and 3.

Information Criteria: Fit Versus Complexity

A model with more parameters will usually fit the observed data better, but this does not necessarily mean that it is the better model.

The **Akaike Information Criterion** balances model fit and complexity:

$$AIC = 2k - 2\ell_{\max},$$

where: k = number of fitted parameters, $\ell_{\max}$ = maximum log-likelihood.

Interpretation

Prefer the model with the lower AIC.

A model with more parameters is preferred only when its improvement in likelihood is large enough to overcome the complexity penalty.

Interactive Exercise: Model Comparison and AIC

Open the interactive demo

Compare two fitted delay models using:

- held-out log-likelihood;
- Exponential versus Gamma distributions;
- AIC and AICc;
- probability predictions for a chosen interval.



https://aisl-at-imperial-college-london.github.io/summer_school_repo/multi_model_test_and_AIC.html

Hypothesis Testing

Hypothesis testing compares data with a default claim.

Null hypothesis

H_0 : no effect, no change, or default value.

Example:

$$H_0 : \mu = 0.$$

Alternative hypothesis

H_1 : there is an effect or difference.

Example:

$$H_1 : \mu \neq 0.$$

We ask: would the observed data be surprising if H_0 were true?

Test Statistic and p-value

A test statistic summarises how far the data are from what H_0 predicts.
For a mean, one common form is

$$z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

p-value

The p-value is the probability, assuming H_0 is true, of seeing a test statistic at least as extreme as the one observed.

A small p-value means the data look unusual under H_0 . It does not measure the probability that H_0 is true.

Errors in Hypothesis Testing

	Truth: H_0 true	Truth: H_0 false
Do not reject H_0	correct	Type II error
Reject H_0	Type I error	correct

Type I error

False positive: we claim an effect when there is none.

Type II error

False negative: we miss an effect that is really there.

In ML, similar trade-offs appear in classification: false alarms versus missed detections.

MLE Versus Unbiased Variance Estimate

For Gaussian data, the MLEs are

$$\hat{\mu}_{\text{MLE}} = \bar{x}, \quad \hat{\sigma}_{\text{MLE}}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

The commonly used sample variance is

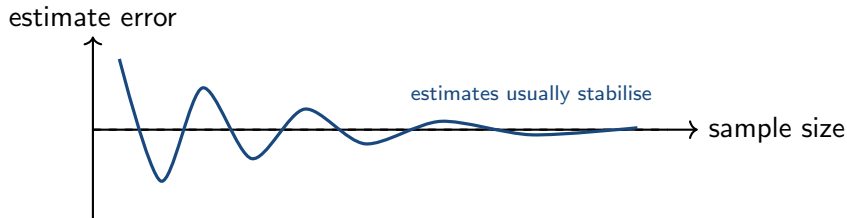
$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

Why two formulas?

The MLE is the parameter value that maximises likelihood. The $n - 1$ formula corrects the typical underestimation of variance caused by estimating $\bar{x}$ from the same data.

What Changes When We Collect More Data?

Parameter estimates vary from sample to sample.

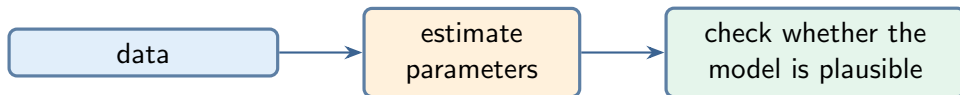


Class exercise

Estimate p , λ , μ , or σ from a small sample and then from a larger sample. Which estimate changes more?

From Estimation to Model Checking

Estimating parameters is not the end of the story.



New question

Suppose we have a new batch of measurements. Is it compatible with a candidate distribution?

This is where p-values can be used as a measure of surprise under a proposed model.

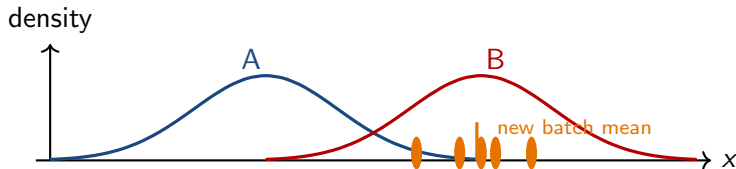
Two Candidate Distributions and a New Batch

Consider two candidate models:

$$A : X \sim \mathcal{N}(0, 1), \quad B : X \sim \mathcal{N}(3, 1).$$

A new batch gives

$$2.1, 2.7, 3.0, 3.2, 3.7, \quad \bar{x} = 2.94.$$



p-test 1: Is the Batch Compatible with Candidate A?

Test the null hypothesis

$$H_0^A : X \sim \mathcal{N}(0, 1).$$

For a batch mean with known $\sigma = 1$,

$$Z_A = \frac{\bar{x} - 0}{1/\sqrt{5}} = \frac{2.94}{0.447} = 6.57.$$

The two-sided p-value is

$$p_A = 2P(Z \geq |6.57|) \approx 4.9 \times 10^{-11}.$$

Interpretation

This batch would be extremely surprising if candidate A were true. At a 5% level, we reject candidate A.

p-test 2: Is the Batch Compatible with Candidate B?

Test the null hypothesis

$$H_0^B : X \sim \mathcal{N}(3, 1).$$

For the same batch mean,

$$Z_B = \frac{\bar{x} - 3}{1/\sqrt{5}} = \frac{2.94 - 3}{0.447} = -0.13.$$

The two-sided p-value is

$$p_B = 2P(Z \geq |-0.13|) \approx 0.89.$$

Interpretation

This batch is not surprising under candidate B. We do not reject candidate B.

How to Interpret the Two p-values

Candidate	p-value	Decision at 5%
$A : \mathcal{N}(0, 1)$	4.9×10^{-11}	reject
$B : \mathcal{N}(3, 1)$	0.89	do not reject

Important warning

A p-value is not the probability that a candidate distribution is true. It measures how surprising the data would be if that candidate were true.

Two p-tests are useful as a compatibility check, but they are not the same as computing $P(A \mid \text{data})$ and $P(B \mid \text{data})$.

Interactive Demo: p-Tests

[Open the interactive p-test demo](https://aisl-at-imperial-college-london.github.io/summer_school_repo/p_test_demo.html)

https://aisl-at-imperial-college-london.github.io/summer_school_repo/p_test_demo.html



Beyond the Mean: Testing Shape and Spread

The previous example tested the batch mean.

Sometimes the mean looks plausible, but the distribution shape does not.

Mean-based test

$\bar{x}$ compared with μ_0

Good for detecting shifts in centre.

Distribution-based test

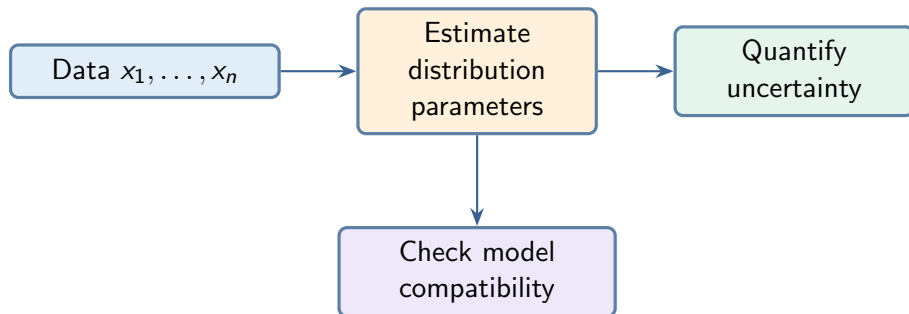
Compare the whole empirical distribution with a candidate CDF.

Examples include Kolmogorov–Smirnov type tests.

Practical message

Choose the test statistic based on what mismatch you care about: centre, spread, tails, or overall shape.

Statistics and Machine Learning: Big Picture



Message

Before supervised learning, statistics teaches us how to estimate unknown quantities and assess whether data are compatible with a proposed distribution.

What to Remember

- Data are samples from a broader population or process.
- Parameters describe distributions; statistics estimate parameters from data.
- Estimates vary from sample to sample.
- Standard errors and confidence intervals quantify uncertainty.
- MLE chooses the parameter values that make the observed data most probable.
- p-values measure surprise under a proposed model, not the probability that the model is true.
- Two p-tests can compare whether a new batch is compatible with two candidate distributions.

Statistics helps us interpret patterns in finite data.

Statistics helps us avoid over-interpreting patterns in finite data.